



SampleAssist®

EMERGENCY CALLOUT MANAGEMENT

CSIT998

PROFESSIONAL CAPSTONE PROJECT

GROUP 14

Work Integrated Learning (WIL) Approach

OUR TEAM - GROUP 14



Rusira Liyanage

Team Leader



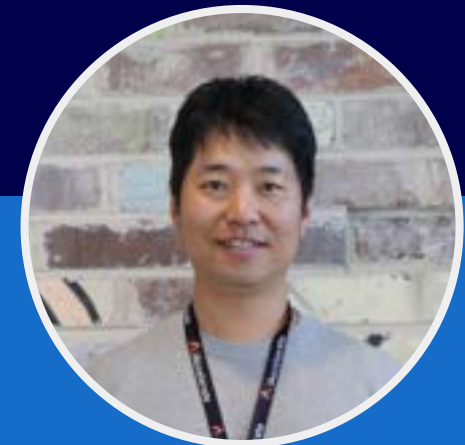
Jinchen Wang

Frontend Lead



Jingwei Li

Software Engineer



Jihoon Lee

AWS Lead



Yunheng Luo

Software Engineer



Le Shi

UI/UX Lead



Yejie Wang

Backend Lead

Background - Pre Employment Drug Testing

- Pre-employment drug testing is vital in high-risk industries (e.g., construction, mining, and transport).
- Drug abuse costs Australia \$6B+ in lost productivity.
(Source: Australian Drug Foundation Report, 2024)
- Desktop-only systems delay emergency response.
- Sample Assist identified the need for mobile accessibility.
- The mobile app enables 24/7 test management and emergency callouts.
- Improves compliance, job tracking, and field decision-making.

Sample Assist (Pty) Ltd

Work Integrated Learning (WIL) Approach



Sample Assist™
www.sampleassist.com

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- MedTec Startup based at IAccelerate, UOW Innovation Campus.
- Pioneers in digital health that specialized in digital medical sample management designed for medical organizations to protect patient privacy.
- Streamlines and manages the collection process, boosting the number of patients that can be assessed each day.
- Enable healthcare centers to reduce administration costs and patient wait times.

The Problem

- Sample Assist's current system operates only via a desktop platform, limiting mobility in urgent or remote situations.

Dependency on a Fixed Location

Desktop-only access restricts off-site personnel and limits real-time emergency management flexibility.

Limited Accessibility During Emergencies

The current desktop platform delays emergency response, increasing risks during incidents.

Delayed Incident Response

The lack of mobility slows medical testing and resource coordination, delaying emergency response and increasing the risk of injury escalation.

Absence of Real Time Communication

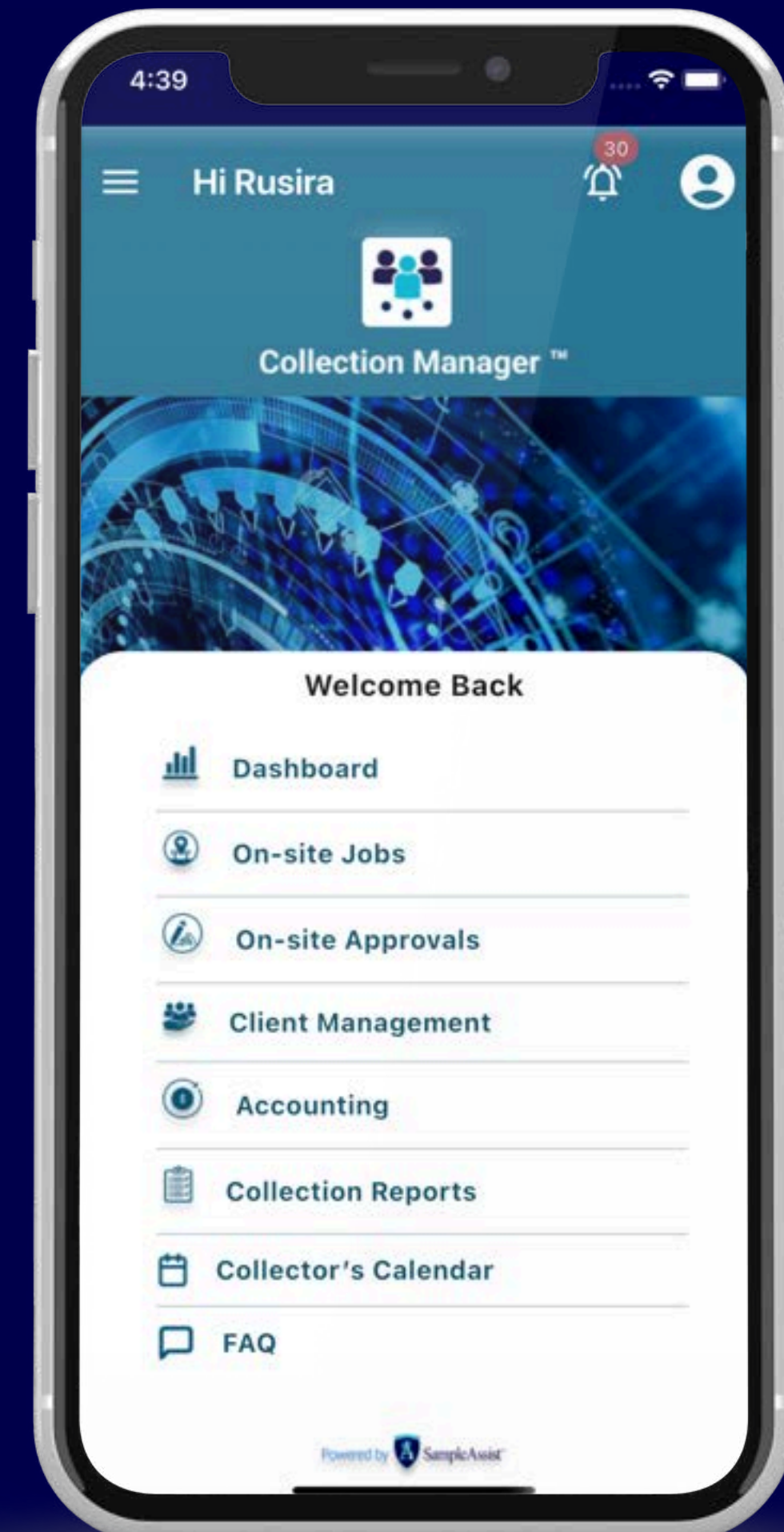
The lack of real-time updates hinders team coordination, slowing decision-making and resource allocation during emergencies.

- Therefore, there is a need for a secure, scalable mobile solution that mirrors essential desktop functionalities.

The Solution

Collection Manager Mobile

- Flutter-based mobile app mirroring core features of Sample Assist's desktop platform.
- Enables real-time job creation, callouts, and client/test handling on the go.
- Cross-platform (iOS & Android) with secure AWS backend integration.
- Field-optimized design for faster, more responsive operations.
- Enhances workflows in high-risk, time-critical scenarios.



The Beneficiaries



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- **Collection Agencies** : Efficient job handling and on-site operations. (key factor)
- **Organization Owners / Managers** : Remote oversight and faster resource allocation.
- **Admin Staff** : Real-time client and job updates.
- **Employers** : Reduced incident downtime and improved safety.
- **Regulators** : Instant access to compliant reports.
- **Employees** : Timely medical response and risk reduction.

Real World Clients



ACCREDITATION NO. 15281

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Advanced Drug Solutions (ADS) (Pty) Ltd



- Drug Testing Agency based in Beresfield, NSW.
- Leading provider of workplace drug and alcohol testing in Australia.
- NATA-accredited and ISO9001 certified, ensuring compliance with national standards.
- Offers on-site and clinic-based testing, training programs, and testing products.
- Serves sectors like transport, construction, and local government.

Relevance of this mobile app for ADS

- Supports real-time job creation and emergency callouts, matching ADS's mobile service model.
- Improves operational efficiency and compliance through seamless data handling.
- Enhances client engagement with timely updates and remote access

Real World Clients

CQ Rescue (CQR) Health Services (Pty) Ltd



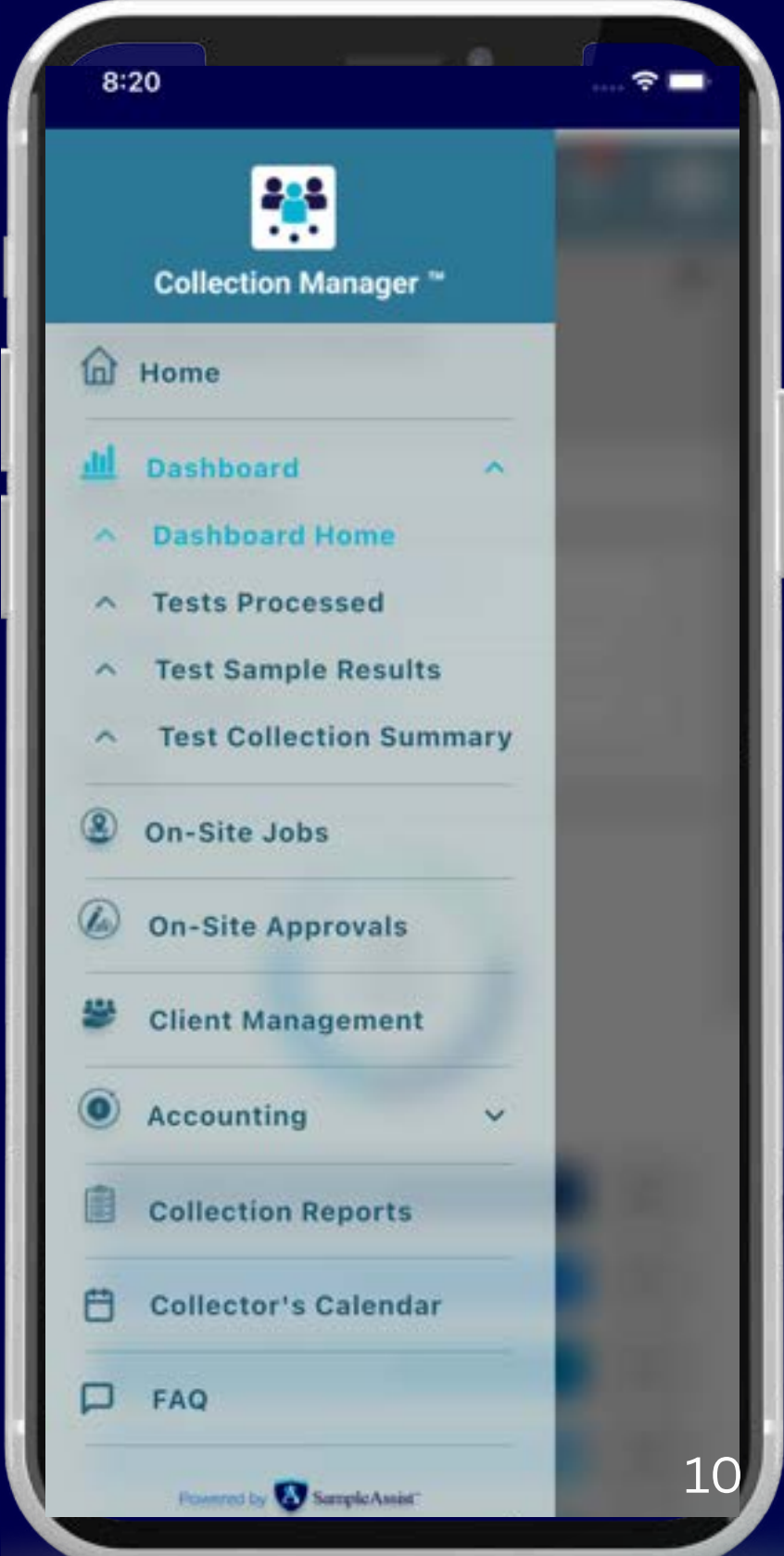
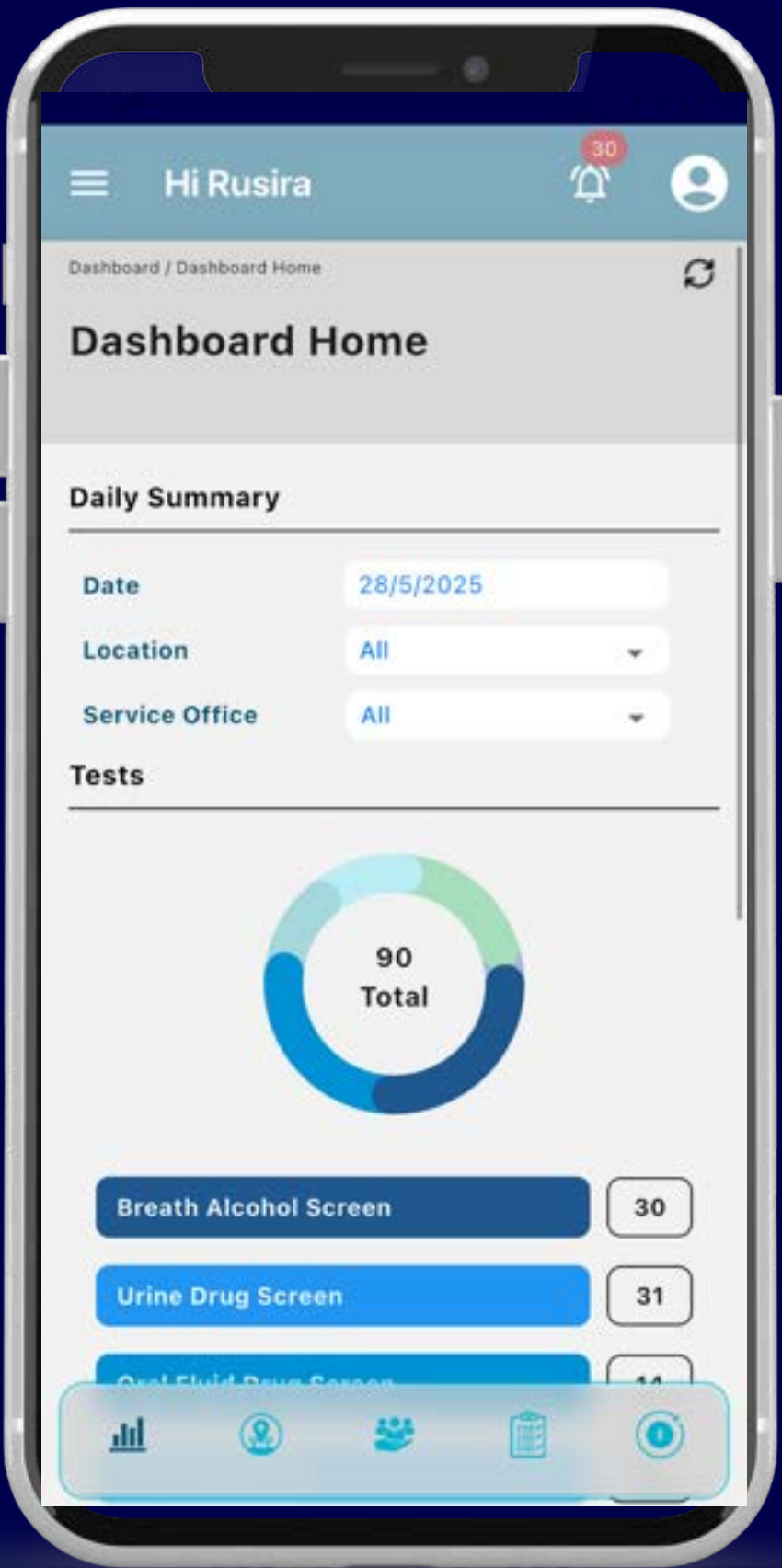
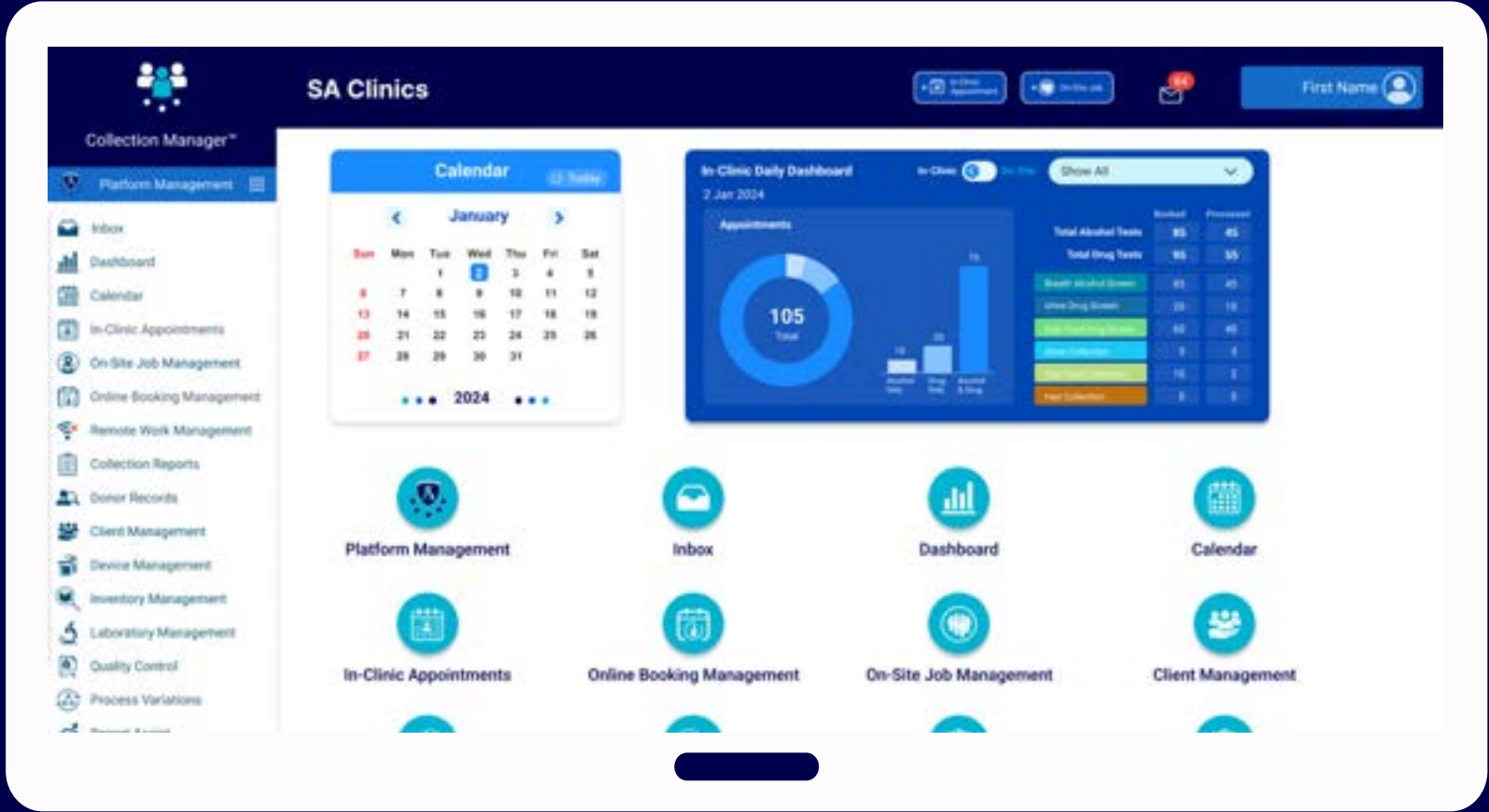
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- Emergency medical and health support provider based in Queensland, Australia.
- Offers 24/7 on-site paramedic and pre-employment medical testing services.
- Supports industries like mining, energy, construction, and rural communities.
- Known for rapid response and remote deployment capabilities.
- Committed to workplace safety and operational continuity.

Relevance of this mobile app for CQR

- Enables rapid job assignment and incident response, vital for emergency callouts.
- Facilitates remote access to test data, crucial for regional and mobile workforces.
- Improves workflow transparency and coordination across field teams.

System Analysis and Design



Design Principles

- 1 Simplicity and clarity
- 2 Style consistency
- 3 System security and user privacy
- 4 Sustainability and Scalability
- 5 Good performance experience



Style Guide



Typography



Bold

Whereas a common understanding of these rights and freedoms is

Medium

Whereas a common understanding of these rights and freedoms is

Regular

Whereas a common understanding of these rights and freedoms is

Light

Whereas a common understanding of these rights and freedoms is

Heading 1 (24px)

Whereas a common understanding of these rights and freedoms is

Heading 2 (20px)

Whereas a common understanding of these rights and freedoms is

Heading 3 (18px)

Whereas a common understanding of these rights and freedoms is

Text (16px)

Whereas a common understanding of these rights and freedoms is

Color

Primary

#151349



Secondary

#01B4D2



Tertiary

#156CC9



Success

#35B700



Info

#1ABCFE



Warning

#FFD52C



Error

#FF3333



Disabled

#D8D8D8



Dis. Button

#879AC1



Requirement Analysis

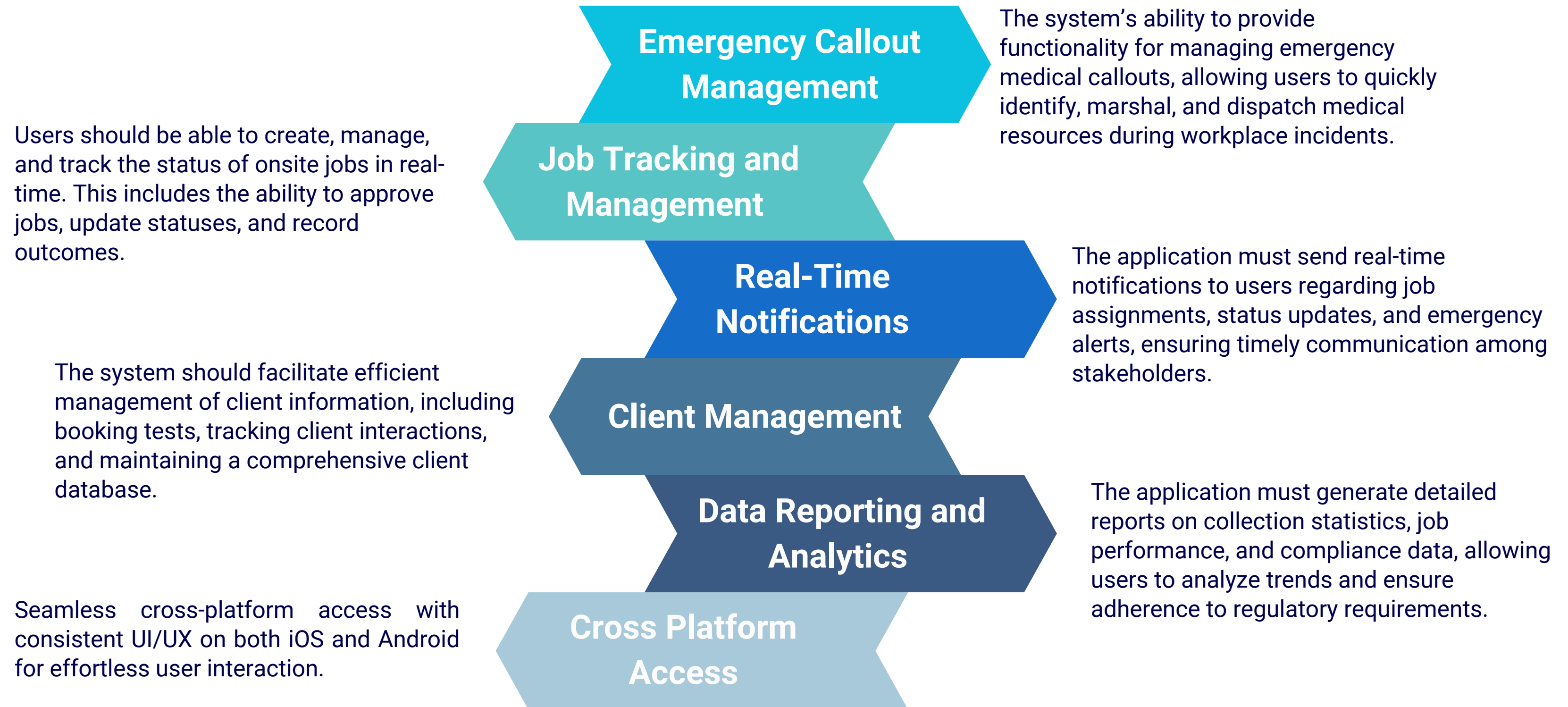
Functional Requirements



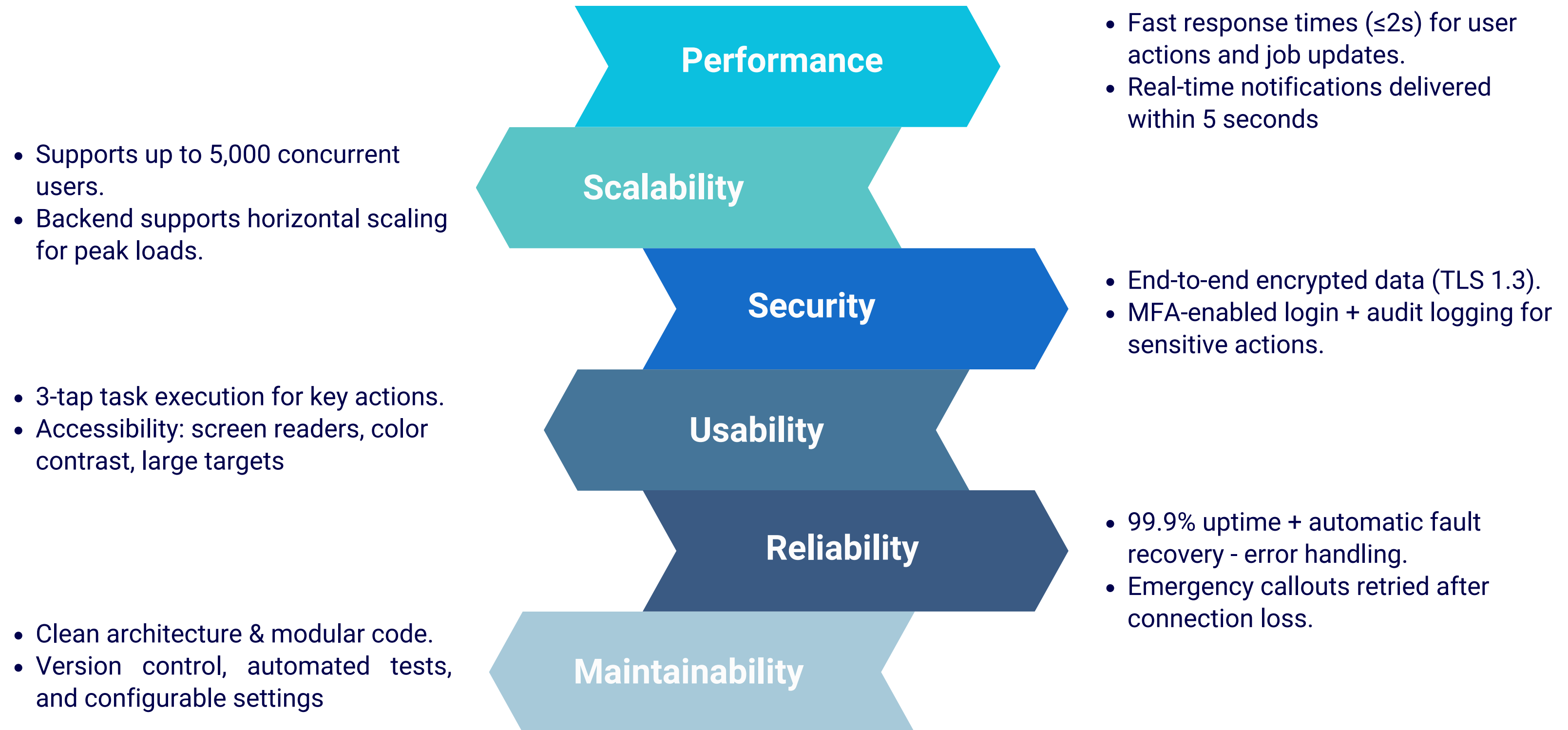
Non-Functional Requirements



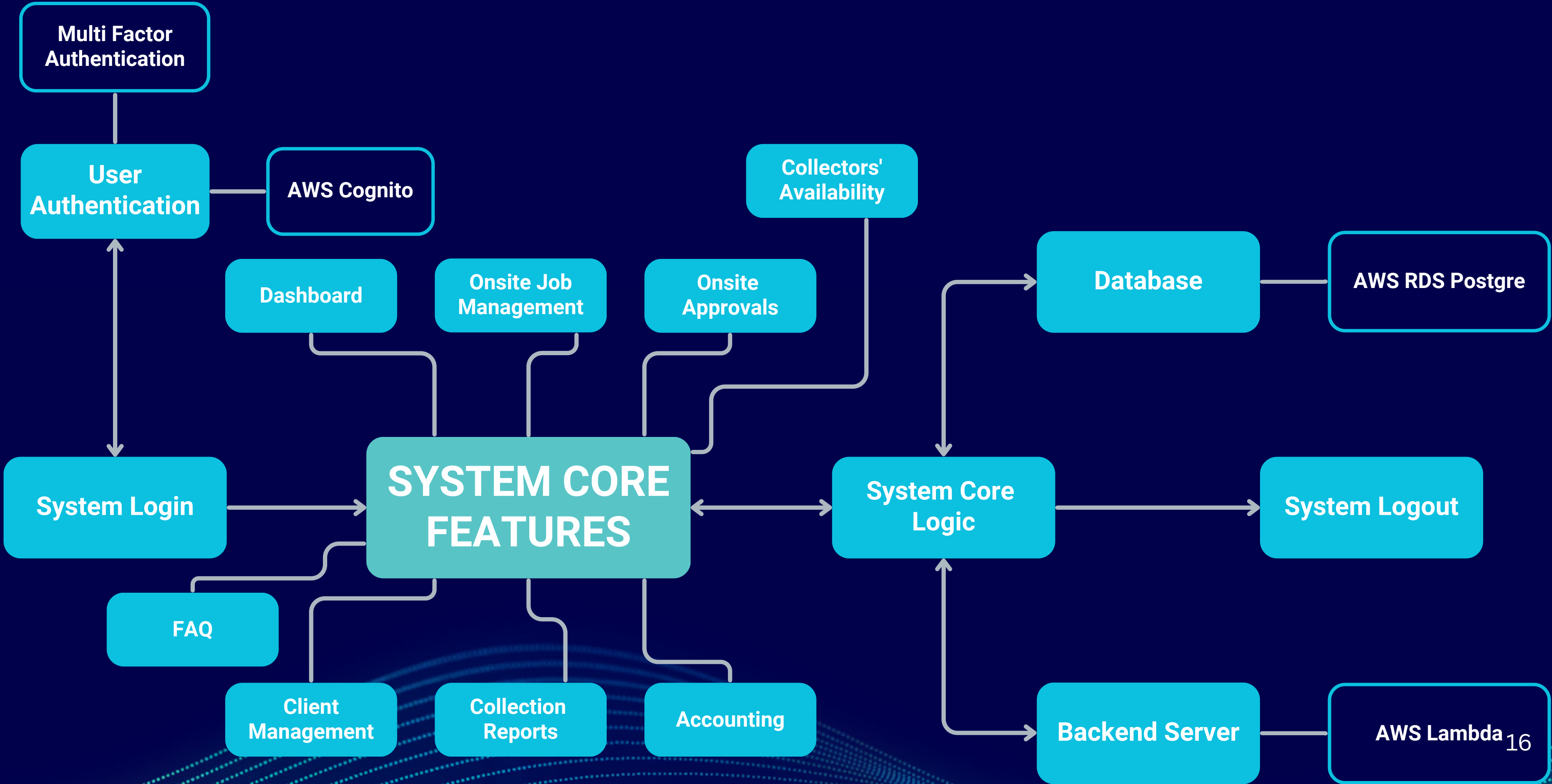
Functional Requirements



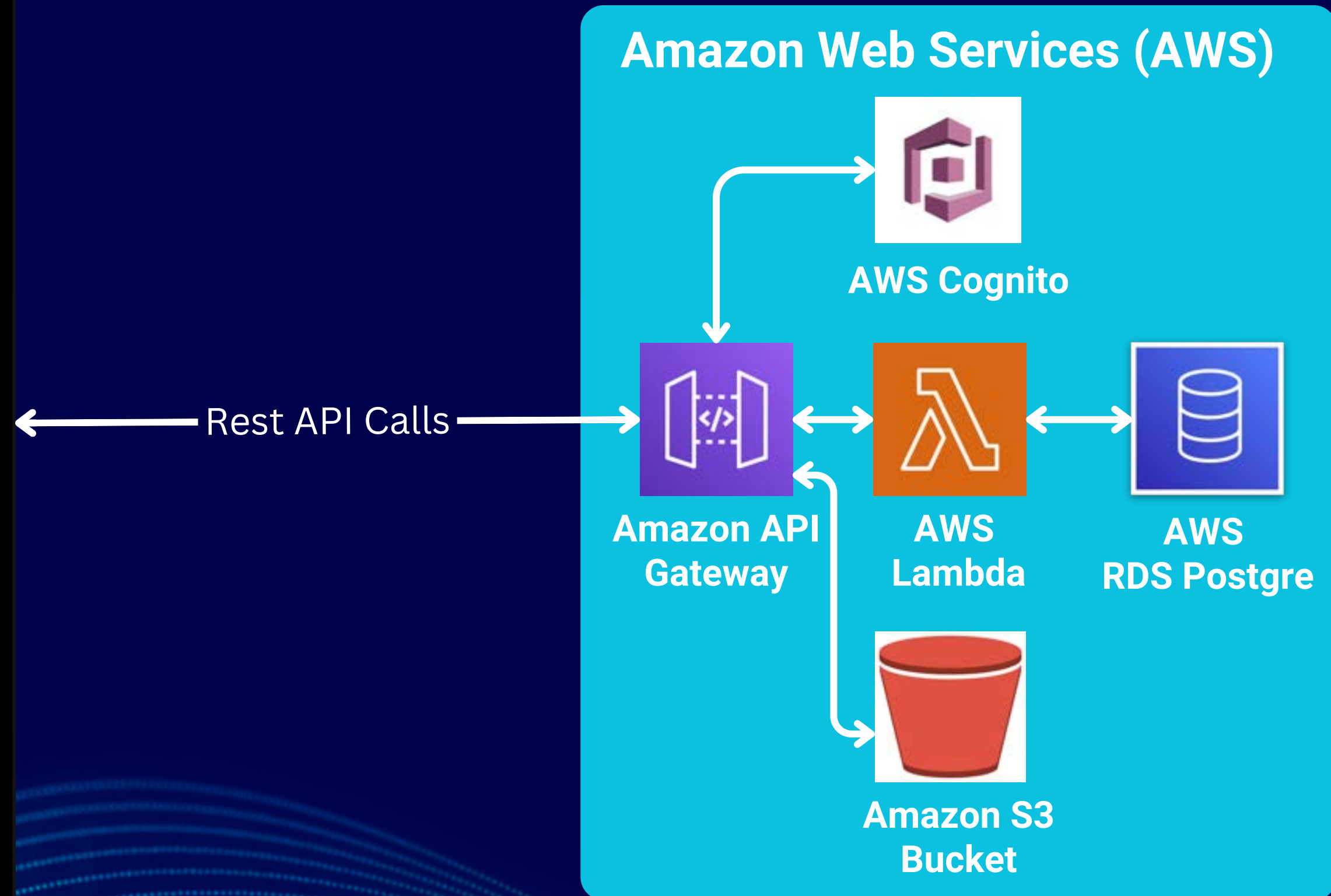
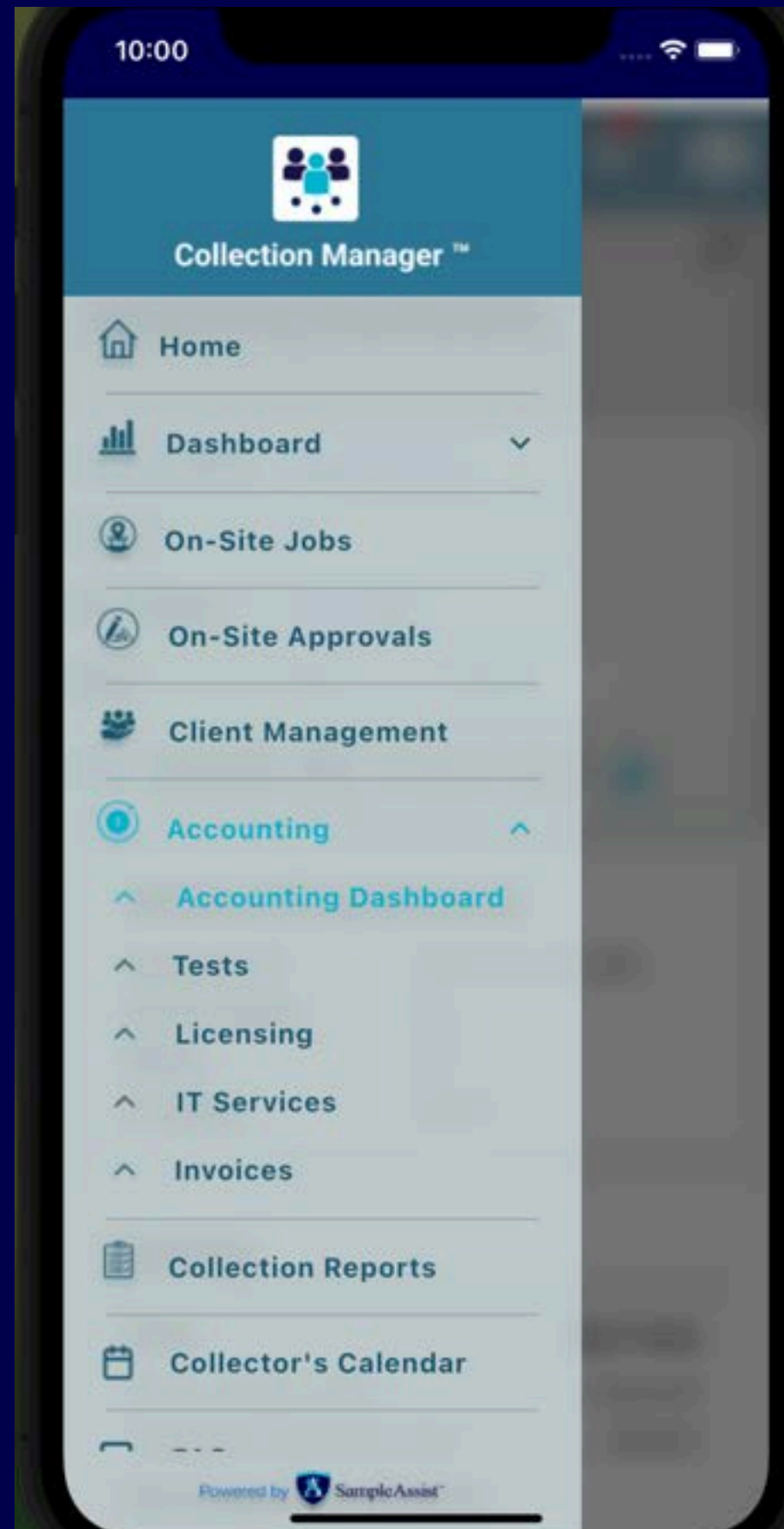
Non Functional Requirements



System Overview



System Architecture



Technology Stack

Frontend

1

 Figma

2

 Flutter

3

 Material UI

3

 pub.dev

Backend

1

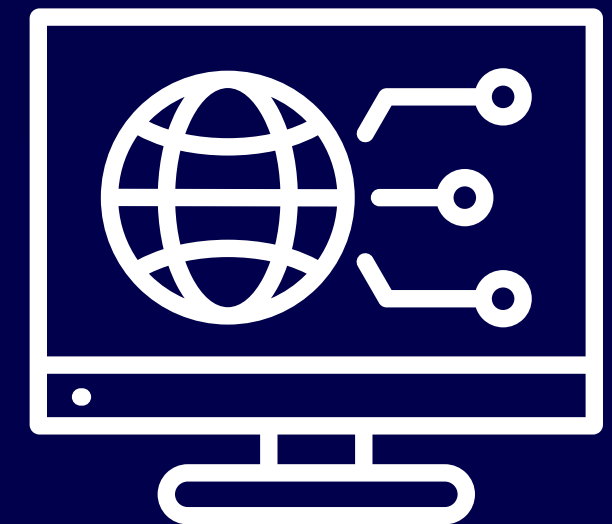
Express JS

2

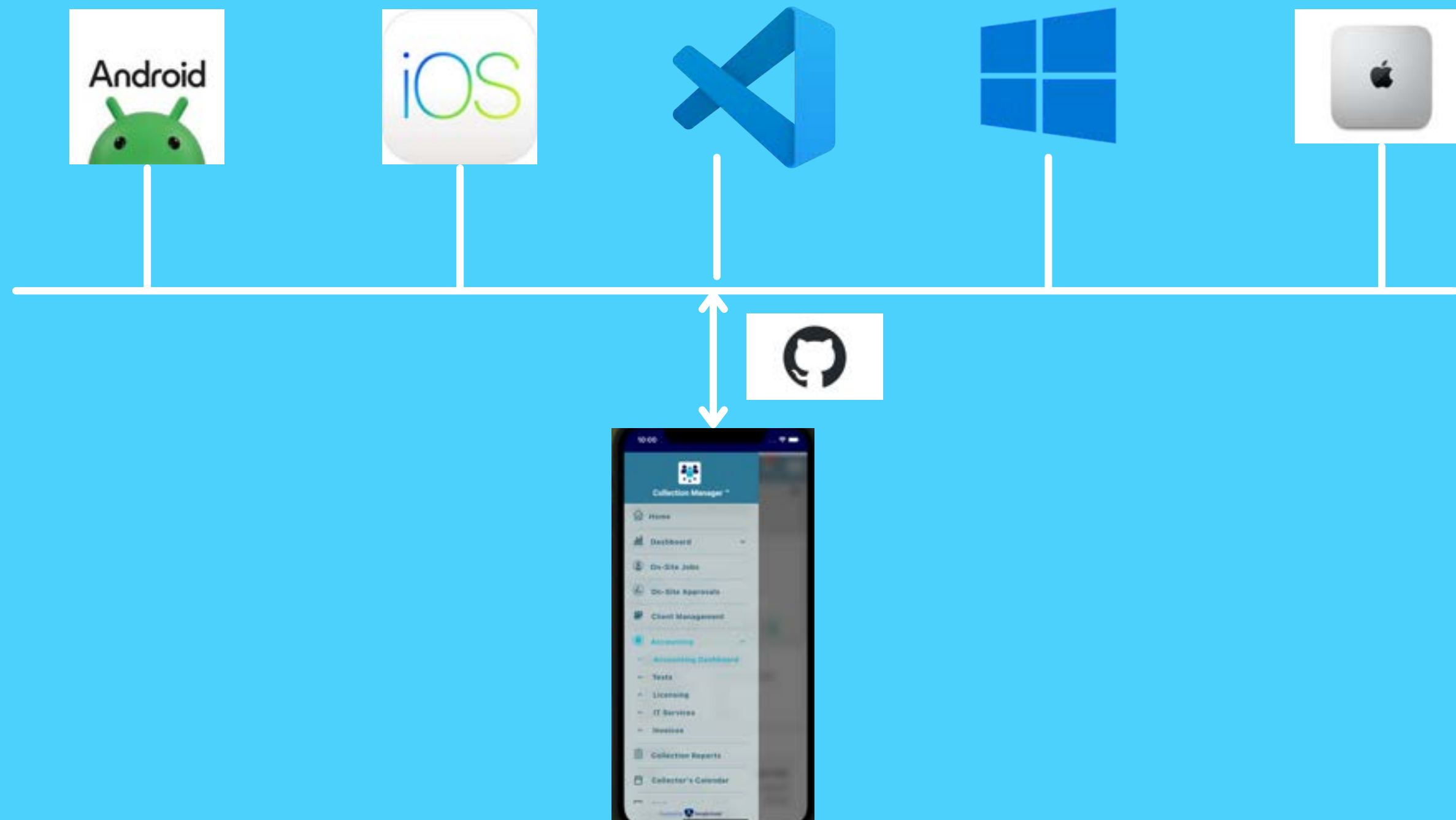

AWS Lambda

3


PostgreSQL



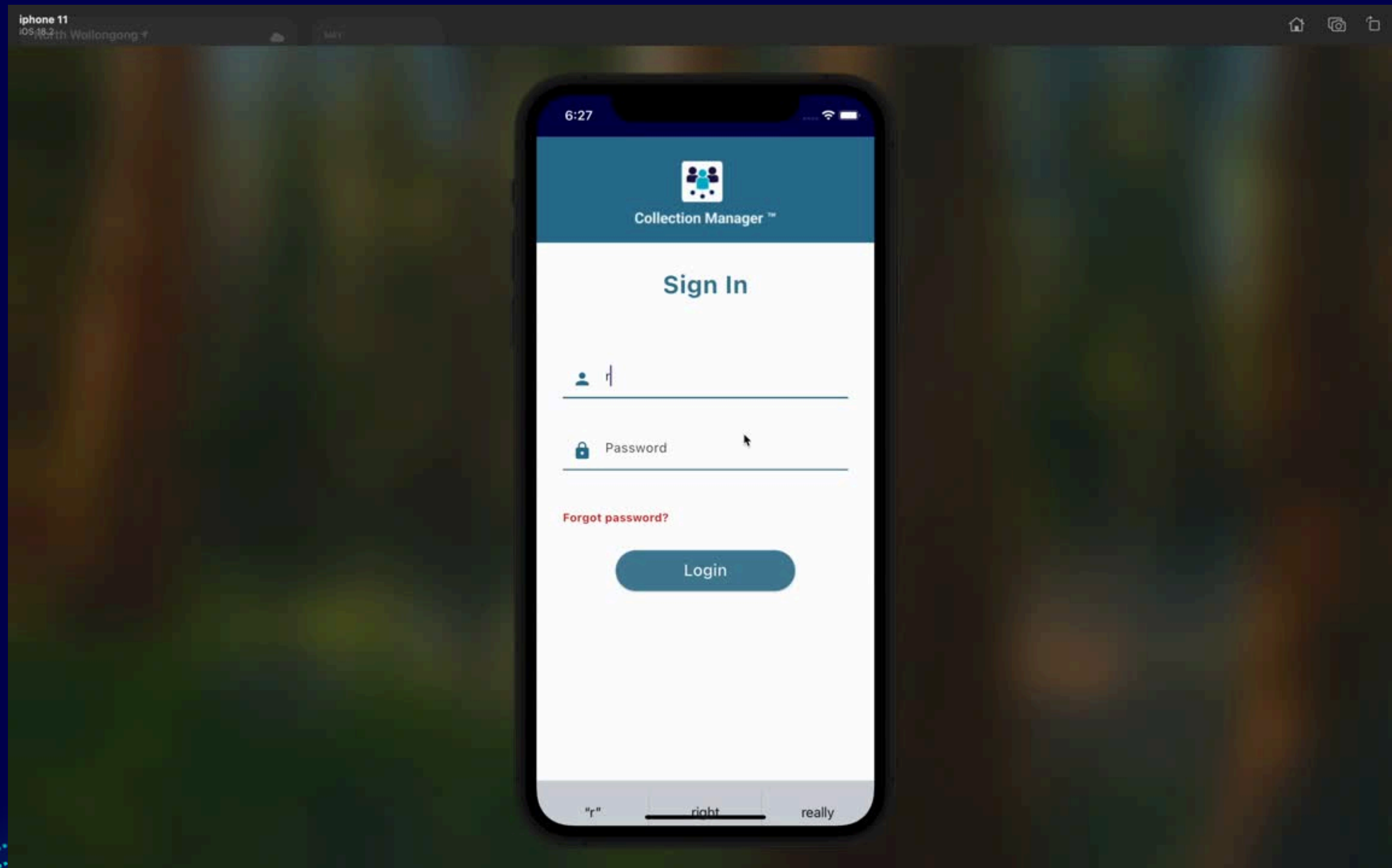
Development Environment



Application Demo

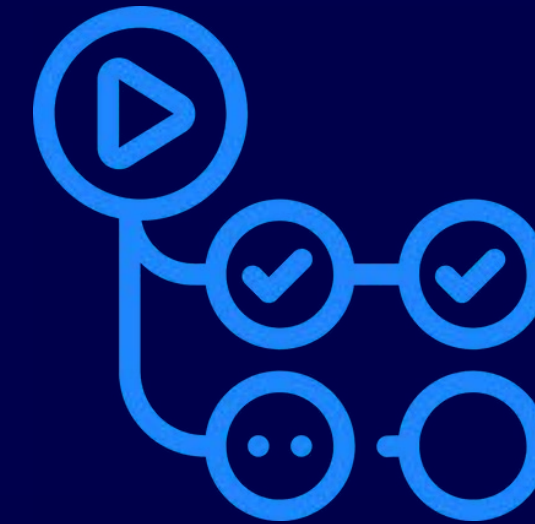
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Iphone 11 XCode Simulator



Future Work

- **Conduct Testing - both iOS and Android**
 - Manual Testing
 - Regression Testing
 - User-acceptance Testing
- **CI/CD Pipeline Completion**
 - Publish into the Apple App Store
 - Publish into the Android Play Store
- **User Feedback Loop**
 - Iteratively improve user experience and add enhancements to meet evolving operational needs.
- **Desktop Feature Migration**
 - As it is required by the collection agencies, the other major functionalities will be considered migrating to the mobile environment in the future.



Thank You!



SampleAssist®

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Contact Us ...

Rusira Senath Liyanage - 8275129
rsipl555@uowmail.edu.au

Jinchen Wang - 8164769
jw447@uowmail.edu.au

Jingwei Li - 8089954
jl225@uowmail.edu.au

Jihoon Lee - 7742873
jl069@uowmail.edu.au

Yunheng Luo - 8539261
yl156@uowmail.edu.au

Le shi - 7890321
ls050@uowmail.edu.au

Yejie Wang - 7598531
yw060@uowmail.edu.au